Inputs

Project Description

```
Arduino project that uses servo motor.
                    ⋮
When the Arduino powers up, it should:
1. Move the servo from 0 to 180 degrees in 1-degree steps
2. When the moving is finished, print the current servo position to the serial monitor
3. Move the servo from 180 to 0 degrees in 1-degree steps
4. When the moving is finished, print the current servo position to the serial monitor
```

Code

```
int pos = 0;
void setup() {
    myservo.attach(9);
}
void loop() {
    for (pos = 0; pos ⩽ 180; pos += 1) {
        myservo.write(pos);
    }
    Serial.println(myservo.read());
    ...
}
```

Test Procedure

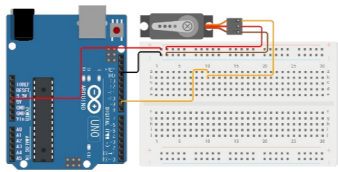
```
System Initializes
→ Sweep starts

First Sweep (0° to 180°)
→ Report position (180°)

Second Sweep (180° to 0°)
→ Report position (0°)
```

Success or **Failure**

Evaluation

Generated Circuit

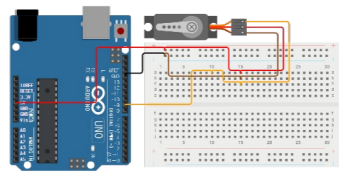
o3-Mini



✅ **Success**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	0
Breadboard Bypass	0

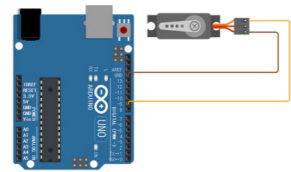
Claude-3.7-Sonnet



❌ **Failure**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	**0**
Breadboard Bypass	**4**

DeepSeek-Coder-v2

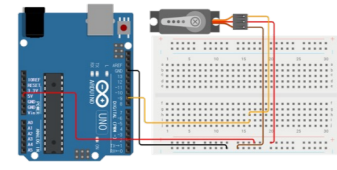


❌ **Failure**

Redundant Connection	0
Extraneous Component	0
Missing Component	**1**
Isolated Component	0
Pin Conflict	**0**
Breadboard Bypass	**2**

Physical Circuit

Reference



✅ **Success**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	0
Breadboard Bypass	0